```
# Installing Python libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')
```

```
# Uploading the Dataset
import pandas as pd
df = pd.read_csv("/content/used_cars_data.csv")
```

```
# To check first five rows
df.head()
```

	S.No.	Name	Location	Year	Kilometers_Driven	Fuel_Type	Transmission	Owner_Type	Mileage	Engine	Power	Seats	New
0	0	Maruti Wagon R LXI CNG	Mumbai	2010	72000	CNG	Manual	First	26.6 km/kg	998 CC	58.16 bhp	5.0	
1	1	Hyundai Creta 1.6 CRDi SX Option	Pune	2015	41000	Diesel	Manual	First	19.67 kmpl	1582 CC	126.2 bhp	5.0	
2	2	Honda Jazz V	Chennai	2011	46000	Petrol	Manual	First	18.2 kmpl	1199 CC	88.7 bhp	5.0	8.(
3	3	Maruti Ertiga VDI	Chennai	2012	87000	Diesel	Manual	First	20.77 kmpl	1248 CC	88.76 bhp	7.0	
4	4	Audi A4 New 2.0 TDI Multitronic	Coimbatore	2013	40670	Diesel	Automatic	Second	15.2 kmpl	1968 CC	140.8 bhp	5.0	

Next steps: Generate code with df | New interactive sheet

```
# To check last five rows
df.tail()
```

	S.No.	Name	Location	Year	Kilometers_Driven	Fuel_Type	Transmission	Owner_Type	Mileage	Engine	Power	Seats
7248	7248	Volkswagen Vento Diesel Trendline	Hyderabad	2011	89411	Diesel	Manual	First	20.54 kmpl	1598 CC	103.6 bhp	5.0
7249	7249	Volkswagen Polo GT TSI	Mumbai	2015	59000	Petrol	Automatic	First	17.21 kmpl	1197 CC	103.6 bhp	5.0
7250	7250	Nissan Micra Diesel XV	Kolkata	2012	28000	Diesel	Manual	First	23.08 kmpl	1461 CC	63.1 bhp	5.0
7251	7251	Volkswagen Polo GT TSI	Pune	2013	52262	Petrol	Automatic	Third	17.2 kmpl	1197 CC	103.6 bhp	5.0
7252	7252	Mercedes-Benz E-Class 2009-2013 E 220 CDI Avan...	Kochi	2014	72443	Diesel	Automatic	First	10.0 kmpl	2148 CC	170 bhp	5.0

```
# This will give us a short information about the Dataset
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7253 entries, 0 to 7252
Data columns (total 14 columns):
 #   Column             Non-Null Count  Dtype  
---  ------             --------------  -----  
 0   S.No.              7253 non-null   int64  
 1   Name               7253 non-null   object 
 2   Location           7253 non-null   object 
 3   Year               7253 non-null   int64  
 4   Kilometers_Driven  7253 non-null   int64  
 5   Fuel_Type          7253 non-null   object 
 6   Transmission       7253 non-null   object 
 7   Owner_Type         7253 non-null   object 
 8   Mileage            7251 non-null   object 
 9   Engine             7207 non-null   object 
 10  Power              7207 non-null   object 
 11  Seats              7200 non-null   float64
 12  New_Price          1006 non-null   object 
 13  Price              6019 non-null   float64
dtypes: float64(2), int64(3), object(9)
memory usage: 793.4+ KB
```

```
# This will help us to read the objects into numerical values
df.describe()
```

	S.No.	Year	Kilometers_Driven	Seats	Price
count	7253.000000	7253.000000	7.253000e+03	7200.000000	6019.000000
mean	3626.000000	2013.365366	5.869906e+04	5.279722	9.479468
std	2093.905084	3.254421	8.442772e+04	0.811660	11.187917
min	0.000000	1996.000000	1.710000e+02	0.000000	0.440000
25%	1813.000000	2011.000000	3.400000e+04	5.000000	3.500000
50%	3626.000000	2014.000000	5.341600e+04	5.000000	5.640000
75%	5439.000000	2016.000000	7.300000e+04	5.000000	9.950000
max	7252.000000	2019.000000	6.500000e+06	10.000000	160.000000

```
df.duplicated().sum()
```

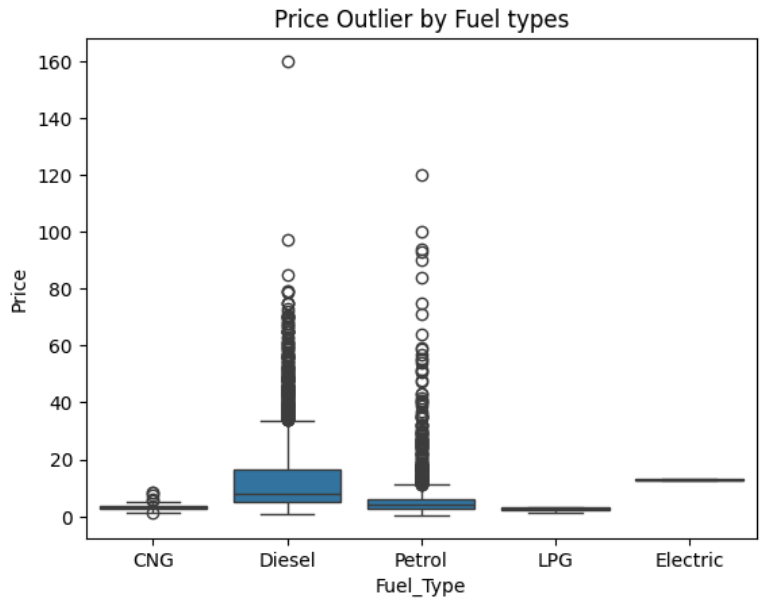
```
np.int64(0)
```

```
df.isnull().sum()
```

	0
S.No.	0
Name	0
Location	0
Year	0
Kilometers_Driven	0
Fuel_Type	0
Transmission	0
Owner_Type	0
Mileage	2
Engine	46
Power	46
Seats	53
New_Price	6247
Price	1234

dtype: int64

```
# Outlier Detection
sns.boxplot(x="Fuel_Type", y="Price", data=df)
plt.title("Price Outlier by Fuel types")
plt.show()
```




```
# Outlier treatment

for col in ['Price', 'Kilometers_Driven']:

    Q1 = df[col].quantile(0.25)

    Q3 = df[col].quantile(0.75)

    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR

    upper = Q3 + 1.5 * IQR

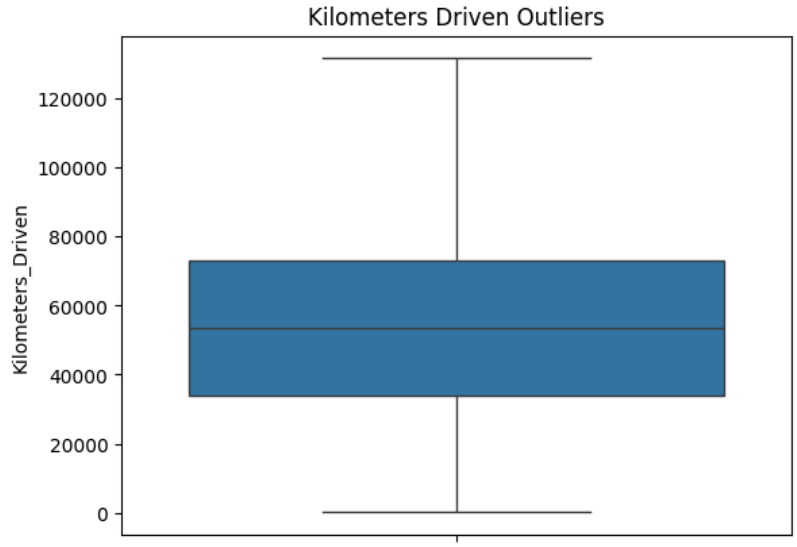
    df[col] = np.where(
        df[col] > upper, upper,
```

```
        np.where(df[col] < lower, lower, df[col])
    )
```

```
# Price outliers
sns.boxplot(y=df['Price'])
plt.title('Price Outliers')
plt.show()

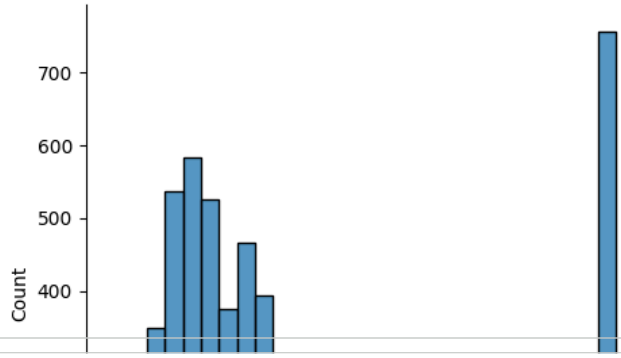
# Kilometers Driven outliers
sns.boxplot(y=df['Kilometers_Driven'])
plt.title('Kilometers Driven Outliers')
plt.show()
```





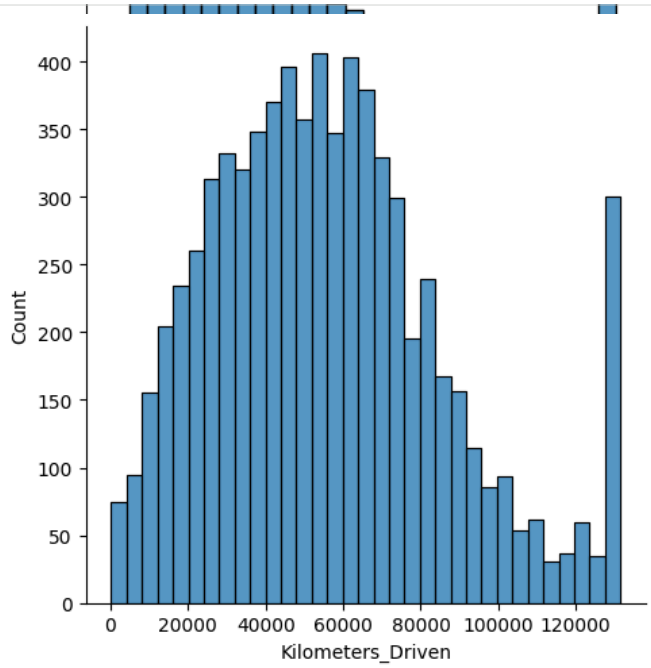

```
# Univariate Analysis
sns.displot(df['Price'])

plt.show()
```




```
# Univariate Analysis
sns.displot(df['Kilometers_Driven'])

plt.show()
```




```
# Bivariate Analysis
sns.regplot(x=df['Price'],y=df['Kilometers_Driven'])


plt.title('Price vs Kilometers Driven')
plt.xlabel('Price')
plt.ylabel('Kilometers Driven')

plt.show()
```

